



Forecasting the Crude Oil prices for last four decades using deep learning approach

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ABSTRACT

A lot of stress is observed on the forecasting of Crude Oil prices due to its liquid nature. Crude Oil is the most liquid commodity in almost all the commodity exchanges around the world. Of late, Artificial Neural Network and most specifically Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) have got a coveted space in the forecasting of asset prices. So, in this piece of research, we optimized the hyperparameters of LSTM and GRU using the well-known Particle Swarm Optimization method to predict the crude oil prices. Thereby, we did a comparative study on which method performed better with an optimized set of hyperparameters. We found that GRU had better performance than LSTM with Root Mean Square Error (RMSE) of 1.23 and an R-squared value of 99.39%.

1. Introduction

Crude Oil is the most important commodity in this civilized world. Being one of the major sources of energy, it has got various uses. Thus a correct prediction of the crude oil prices is necessary for economic world as well as for governments of several countries. A large number of researches have already been engaged in the prediction of crude oil prices. These ways comprise fundamental, technical, and quantitative methods. It is worth mentioning here that the fundamental methods consist of the prediction of crude oil prices based on supply–demand, geo-political issues, economy of the countries, etc. The technical methods of prediction consist of studying the charts and the price action of crude oil. The quantitative methods of prediction consist mainly of studying the time series data and generating mathematical models based upon this. So in this work, we take up the task of predicting crude oil prices using quantitative methods.

Recently, [Busari and Lim \(2021\)](#) devised a forecasting methodology for forecasting the prices of Crude Oil. They devised an ensemble model using GRU and LSTM. They found that the model Adaboost-GRU has performed better compared to Adaboost-LSTM. The benchmark, in this case, was the simple LSTM and GRU methods. [Sun et al. \(2018\)](#) proposed an **interval decomposition ensemble (IDE)** learning approach to forecast interval-valued crude oil price by integrating **bivariate empirical mode decomposition (BEMD)**, **interval MLP (MLP)**, and **interval exponential smoothing method (Holt)**. They

found that their models perform significantly better than other benchmark models. [Ghaffari and Zare \(2009\)](#) predicted the daily variation of the crude oil price of the West Texas Intermediate (WTI). They used a data filtering algorithm to reduce the effect of unpredictable short-term disturbances. [Gumus and Kiran \(2017\)](#) discovered the factors affecting the crude oil prices using XGBoost. They thereby forecasted crude oil prices. [Jammazi and Aloui \(2012a\)](#) used the **multilayer back propagation neural network and Harr A trous wavelet decomposition** to predict crude oil prices. [Li et al. \(2019\)](#) proposed a forecasting method based on **online media text mining**, with the aim of capturing the more immediate market antecedents of price fluctuations. [Miao et al. \(2017\)](#) considered six factors-supply, demand, financial market, commodities market, speculative, and geopolitical; to test their significance in estimating various forecasting models. They found out that the **Least Absolute Shrinkage and Selection Operator (LASSO)** regression method is the best in providing significant improvements in the forecasting accuracy. They then studied the time-varying nature of the relationship between factors and oil prices to explain recent movements in crude oil prices.

[Orojo et al. \(2019\)](#) forecasted oil prices using **Multi-recurrent Network (MRN)**. [Gabralla and Abraham \(2013\)](#) made a comprehensive survey covering the methods, results and experiments for predicting oil prices. [Wang et al. \(2018\)](#) proposed a **semi-heterogeneous** approach to combine crude oil price forecasts. In this approach, a set of

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decomposition methods interacts with a set of forecasting techniques. The approach first decomposes the original price series using four decomposition methods, such as **Wavelet Analysis, Singular Spectral Analysis, Empirical Mode Decomposition, and Variational Mode Decomposition**. Then it uses four different forecasting techniques, such as **Autoregressive Models, Autoregressive Integrated Moving Average Models, Artificial Neural Networks, and Support Vector Regression Models**, to forecast the components from each decomposition method. In the end, the approach reconstructs the price forecasts from the forecasted components. In this process, sixteen price forecasts in total for combination are generated. Wang et al. (2020) proposed a **multi-granularity heterogeneous combination** approach to predict crude oil prices. They used different feature selection techniques like a **filter, wrapper, and embedded methods**. Thereby, they used three different forecasting models: **Linear regression (LR), Artificial neural network (ANN), and Support Vector Machine (SVR)**. In the end, it is optimized by **Artificial Bee Colony (ABC)** techniques. Wu et al. (2019) devised a novel model based on **Ensemble Empirical Mode Decomposition (EEMD) and Long Short-term Memory (LSTM)**. Xie et al. (2006a) proposed a new method for crude oil price forecasting based on a **Support Vector Machine (SVM)** and compared the performance of the proposed method with **ARIMA and BPNN**. They found that the proposed method performed better.

Yu et al. (2022c) proposed a **novel compressed sensing-based learning paradigm by integrating compressed sensing-based denoising (CSD)** and other powerful AI tools. The reason for passing through CSD is to remove the noise present in the data. They found that their proposed model performed better than other benchmark models. Yu et al. (2022b) improved the existing decomposition-ensemble techniques. In this model, all the decomposed modes are thoroughly analyzed to find out hidden information and are then reconstructed to provide valuable information. They found that their proposed model statistically outperforms all benchmark models. Yu et al. (2022a) devised a novel model **integrating ensemble learning paradigm integrating ensemble empirical mode decomposition (EEMD) and extended extreme learning machine (EELM)**. The integration is done using a simple addition ensemble method.

Yu et al. (2017) studied the importance of **Support Vector Machine (SVM)** against five other methods such as **feed-forward neural networks (FNN), auto-regressive integrated moving average (ARIMA) model, fractional integrated ARIMA (ARFIMA) model, Markov-switching ARFIMA (MS-ARFIMA) model, and random walk (RW) model** and found that **SVM** is better than all of them. Zhang et al. (2022) ensemble empirical mode decomposition (EEMD) method to decompose the crude oil prices data. Thereafter, they used the **least square support vector machine together with the particle swarm optimization (LSSVM - PSO) method and the generalized autoregressive conditional heteroskedasticity (GARCH) model** to forecast the data component-wise. In the end, the forecasted data from each component are added to forecast the data. Zhao et al. (2017) devised a **deep learning ensemble approach** to predict crude oil prices. They used **Stacked Denoising Autoencoder (SDAE) model and bagging**. Their proposed method takes the merit from the two methods and produces better results when compared to other benchmark models.

The present work provides a detailed analysis of the use of deep learning models for forecasting crude oil prices. Crude oil prices are highly volatile and subject to various factors such as geopolitical events, supply and demand, and economic conditions. Accurate forecasting of crude oil prices is crucial for decision-making in the energy industry, as it can help companies plan their investments, manage risks, and optimize their operations.

In the field of predicting crude oil prices, numerous research studies have been conducted to investigate the predictive power of deep learning algorithms. However, it has been observed that very few studies have focused on exploring the effectiveness of Gated Recurrent Units (GRU) in this domain. Furthermore, although Long Short-Term Memory

(LSTM) and GRU are widely used deep learning models, Particle Swarm Optimization (PSO) has not been extensively utilized to optimize their hyperparameters for crude oil price prediction. Particularly, the effect of PSO on optimizing the hyperparameters of GRU in predicting crude oil prices has not been explored much in the literature. Moreover, no research has been conducted on such a vast dataset spanning from 1983 to 2021 to predict crude oil prices.

Therefore, our study aims to address these research gaps by investigating the effectiveness of LSTM and GRU in predicting crude oil prices. A comparative study is conducted on the performance of LSTM and GRU models in forecasting crude oil prices. Additionally, we aim to explore the potential benefits of using PSO to optimize the hyperparameters of LSTM and GRU for crude oil price prediction. Furthermore, our study intends to use a large dataset covering almost four decades to obtain a more comprehensive understanding of the predictive power of deep learning models in this domain. The hyperparameters in these models using Particle Swarm Optimization (PSO) technique are optimized. PSO is a metaheuristic optimization algorithm that mimics the behavior of a swarm of particles moving in search of an optimal solution.

In Section 2, we will give a brief description of the data used, and the analyzing tools. Section 3 deals with results and discussion and Section 4 discusses the conclusions.

2. Methodology

2.1. Description of analyzing tools

2.1.1. Data description

The data has been collected from [investing.com](https://www.investing.com) (Investing.com, 2022). The original dataset is from 31 March 1983 to 10 September 2021. The data contains the daily closing prices of Crude Oil. The graph of the daily closing prices of crude oil from 1 March 1983 to 10 September 2021 has been shown in the Fig. 1. The graph of daily logarithmic returns from 1 March 1983 to 10 September 2021 has been shown in Fig. 2.

Some important terms used in this model are mentioned and described below.

1. **Forget Gate** This gate concatenates the output from the last module (h_{t-1}) and the input of the present module (x_t), and passes it through a sigmoid gate.
2. **Sigmoid Gate** A sigmoid gate produces an output value between 0 and 1. It specifies how much information should be transmitted. A value of 0 means "Nothing should get through", while a value of 1 means "Everything should get through".
3. **Cell State** The cell state is similar to a conveyor belt. It follows the entire chain, with only a few minor linear interactions. It is very simple for information to simply flow along it unchanged.
4. **tanh gate** A sigmoid gate produces an output value between -1 and 1. Tanh gate is just one of the gates that can provide nonlinearity, allowing the neural network as a whole to be used for universal approximation.
5. **Reset Gate** The Reset Gate is in charge of the GRU network's short-term memory. It regulates the flow of information out of memory.
6. **Update Gate** The update gate is critical in determining the amount of information to be passed along in the future. The update gate regulates the flow of data into memory.
7. **Hadamard's way of multiplication** Hadamard product of two matrices is the matrix whose entries are the products of the corresponding entries of the input matrices.

A brief description of the statistics of the data from the above graph has been computed as in Table 1.



Fig. 1. Daily closing prices of crude oil from 1983 to 2021.

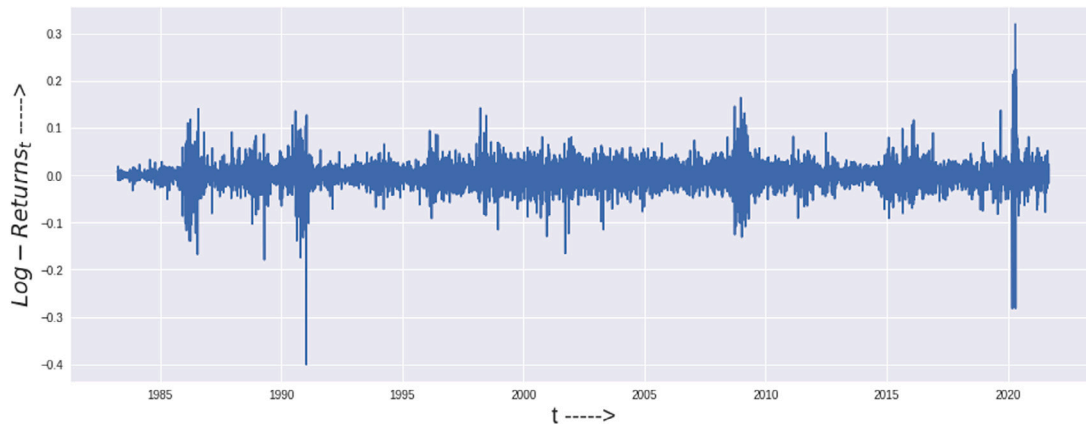


Fig. 2. Daily change in percentage of the closing prices of crude oil from 1983 to 2021.

Table 1
Statistics on crude oil data.

Statistical metrics	Values
Count	9748
Mean	43.58
Standard deviation	27.95
Minimum	-37.63
25%	20.21
50%	30.85
75%	61.50
Maximum	145.29

2.1.2. Long Short-Term Memory (LSTM) method

The LSTM is a forecasting technique based on Neural Networks. Hochreiter and Schmidhuber (Colah.github.io, 2021b) devised it in 1997. A sample LSTM structure is shown in Fig. 3. The main advantage of LSTM technique over simple Neural Networks is its ability to retain information for a longer period of time. Basically, LSTM is having a chain of repeating modules where each module has four neural networks. In the Fig. 3, we can see a horizontal line on the top. It is known as the “Cell State”. The cell state is like a conveyor belt that passes through each module with minor modifications.

Steps used in LSTM method

1. In the first step, we decide the amount of information that is to be passed on to the next module and the amount of information that is to be retained. For that, the concept of forget gate is introduced. This gate concatenates the output from the last module (h_{t-1}) and the input of the present module (x_t), and

passes it through a sigmoid gate. The output from the forget gate is a value between ‘0’ and ‘1’, where ‘0’ means no information is to be retained and ‘1’ means complete information is to be retained.

The equation for this step is given below.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (1)$$

where, f_t = output from the forget gate.

W_f = Weights applied to the forget gate.

b_f = Bias term applied to the forget gate.

2. In the next step, it is to be decided which information and what information are to be added to the cell state. The concatenation of h_{t-1} and x_t is passed through a sigmoid gate (σ) and a tanh gate. The sigmoid gate, known as the “Input Gate Layer” decides which information is to be added. The tanh gate outputs a vector of values that is to be added to the cell state. The equations are given below.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (2)$$

$$X_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (3)$$

where, i_t = output from the Input Gate Layer.

W_i = Weights applied to the Input Gate Layer.

b_i = Bias term applied to the Input Gate Layer.

X_t = output from the tanh Gate Layer.

W_c = Weights applied to the tanh Gate Layer.

b_c = Bias term applied to the tanh Gate Layer.

3. The i_t and the X_t calculated from Eqs. (2) and (3), are to be multiplied in Hadamard’s way of multiplication. The result from

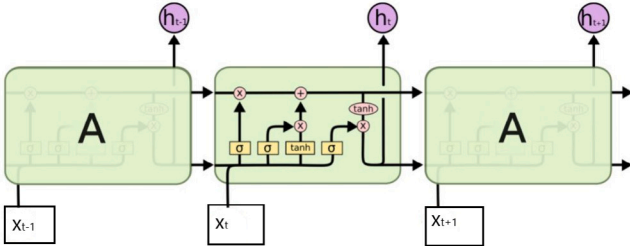


Fig. 3. A sample LSTM diagram (Colah.github.io, 2021a).

this multiplication is added to the Hadamard product of f_t and C_{t-1} to compute C_t .

The equation is given below.

$$C_t = f_t * C_{t-1} + i_t * X_t \quad (4)$$

where, C_{t-1} = Old value of Cell State.

C_t = New value of Cell State.

- In the last step, the concatenation of the h_{t-1} and x_t is passed through a sigmoid gate. And the cell state information is passed through a tanh gate to get the output between -1 and $+1$. The Hadamard's product of both the resultants is then computed to get the desired output.

The equations are given below.

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (5)$$

$$h_t = o_t * \tanh(C_t) \quad (6)$$

where, o_t = output from the last sigmoid gate of the current module.

W_i = Weights applied to the last sigmoid gate of the current module.

b_i = Bias term applied to the last sigmoid gate of the current module.

h_t = output from the current module.

2.1.3. Gated Recurrent Unit (GRU)

GRU was first introduced by [Cho et al. \(2014\)](#) in 2014. A sample GRU structure is shown in [Fig. 4](#). GRU comes as a solution to the vanishing gradient problem of Recurrent Neural Networks (RNN). GRU consists of an update gate and a reset gate. These are used to decide or control which information is to be passed to the output. These two gates can retain the information for a long time if desired.

The stages in GRU are given below.

- In the first step, we compute the value through the update gate and the reset gate. Multiplying the input x_t with its own weight $W^{(z)}$ and, the output from the previous module h_{t-1} with its own weight $U^{(z)}$ and then adding both the results through a sigmoid gate to compute the update gate resultant. The update gate is crucial in its role to determine the amount of information to be passed along towards future.

The equation for the above process is given as follows.

$$z_t = \sigma(W^{(z)}x_t + U^{(z)}h_{t-1}) \quad (7)$$

where, $W^{(z)}$ = Weights associated with the input.

$U^{(z)}$ = Weights associated with the output from the previous module.

- In the next step, the reset gate value is calculated. This step is similar to that of the update gate. Reset Gate basically gives us an idea about how much of the past information is to be forgotten. Here we multiply x_t and h_{t-1} with their respective weights, add it and pass it through a sigmoid gate.

The equation is given below.

$$r_t = \sigma(W^{(r)}x_t + U^{(r)}h_{t-1}) \quad (8)$$

where, $W^{(r)}$ = Weights associated with the input.

$U^{(r)}$ = Weights associated with the output from the previous module.

- In this step, we multiply x_t with weight W and, h_{t-1} with weight U . Thereafter, the resultant from Hadamard's product of $U h_{t-1}$ and r_t is added to $W x_t$. After addition, the added result is then passed through a tanh gate.

The equation is given below.

$$h'_t = \tanh(W x_t + r_t \odot U h_{t-1}) \quad (9)$$

where, W = Weights associated with the input.

U = Weights associated with the output from the previous module.

- In the final step, we need to calculate the output h_t from this module. At first we compute the Hadamard product of z_t and h_{t-1} . Then, we compute the Hadamard product of $(1 - z_t)$ and h'_t . Thereafter, we add the results from the above two products. The equation for this situation is given below.

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot h'_t \quad (10)$$

2.1.4. Particle Swarm Optimization (PSO)

PSO was developed by Kennedy and Eberhart in 1995 ([Kennedy and Eberhart, 1995](#)). It was developed by imitating the social behavior of a flock of birds in their quest for food. The main objective of the PSO is to optimize a cost function. This cost function depends on the proposed optimization formulation of the function $f(X)$. Here, $X = (x_1, x_2, \dots, x_n)$ is a vector, and $f(X)$ is the fitness function (or objective function). The steps involved in the process are explained below.

- In the beginning, we create a population of events, which is uniformly distributed over X .
- We calculate the position of each particle as per the fitness function (objective function). That is, the value of the position of each particle is nothing but the value of the fitness function of that particle.
- If the present position of the particle is better than the previous best position, then we update the best position of the particle.
- Thus the best particle is chosen as per the particle's last best positions.
- The velocities of the particles are then updated by the following equation.

$$V_i^{t+1} = W \cdot V_i^t + c_1 U_1^t (P_{b_i}^t - P_i^t) + c_2 U_2^t (g_b^t - P_i^t) \quad (11)$$

- In the next step the particles are moved to their new positions according to the following relations.

$$P_i^{t+1} = P_i^t + V_i^{t+1} \quad (12)$$

- Step (b) will be repeated till the stopping criteria is fulfilled.

2.1.5. Goodness of fit

The goodness-of-fit is a statistical test that determines how well sample data fits a distribution from a population that is from a normal distribution. In simpler words, goodness-of-fit tests make inferences about observed values. In this work, we will use the chi-square test, which is the most common among other goodness-of-fit methods.

The chi-square test is given as:

$$\chi^2 = \sum_{i=1}^n \frac{(x_i - \hat{x}_i)^2}{\hat{x}_i}$$

where, x_i = Observed Value

$\hat{x}_i$ = Expected Value

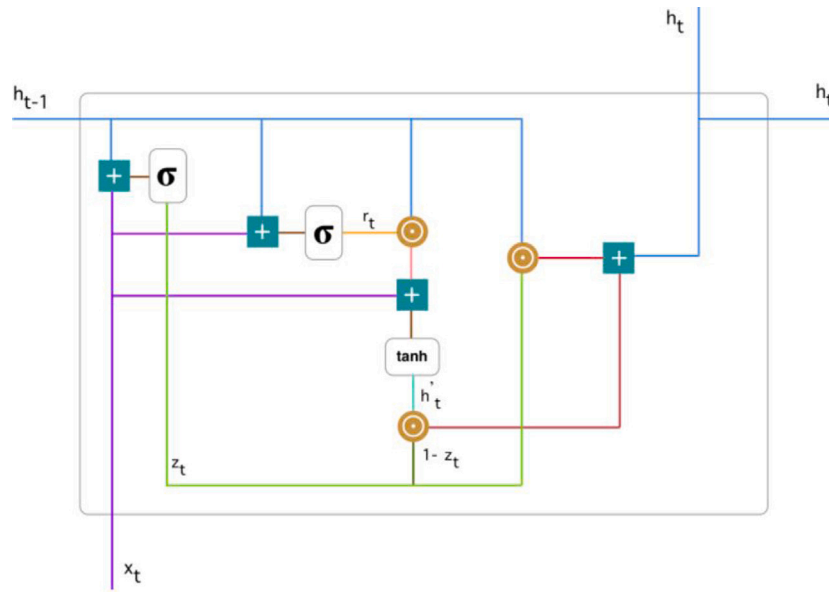


Fig. 4. A sample GRU diagram (Medium, 2022).

2.1.6. Auto-correlation for error terms

To check the degree of similarity between the error terms and their lagged data over successive time intervals; the autocorrelation test and the Durbin-Watson test are performed.

2.1.7. Trend analysis

In order to check for the presence of any trends in the data, the “pandas” package of Python is used. Most Time Series analyses consist of a number of elements. In order to deconstruct our observation into various components, each of which represents one of the underlying categories of patterns, we must now apply decomposition techniques.

Several models should be taken into account while decomposing time series data.

1. Additive Model

When the fluctuations around the trend do not change with the level of the time series, this model is employed. Here, a time series' constituent parts are simply added using the following formula:

$$y(t) = \text{Level}(t) + \text{Trend}(t) + \text{Seasonality}(t) + \text{Noise}(t)$$

2. Multiplicative Model

If the trend is proportionate to the level of the time series, it is employed. Here, a time series' constituent parts are simply multiplied by one another using the following formula:

$$y(t) = \text{Level}(t) * \text{Trend}(t) * \text{Seasonality}(t) * \text{Noise}(t)$$

2.1.8. Hyper parameters to be optimized

Hyper parameter is basically a parameter whose value is used to control the learning process of the algorithm. In this work, the following hyper parameters will be optimized to get the best results.

- Gradient Descent Algorithm** The role of Gradient Descent Algorithm is to minimize the cost function in various machine learning algorithms. In our work between Nadam and Adam.
- Activation Function** It is basically used to determine whether a neuron is to be activated or not. It is mainly used to ensure non-linearity to the process. In our research, we try to optimize between ‘tanh’ and ‘sigmoid’ activation functions.
- Loss** It is used to compute the prediction error of the neural networks.
- Batch Size** It gives us the number of samples to work before updating the model.
- Epochs** It gives us the information of the number of passes of the training set which the algorithm has already completed.

2.2. Score metrics

To compute the accuracy of our results, we use the R-squared value, Mean Absolute Error, Root Mean Square Error, Correlation Coefficient, Relative Absolute Error, and Root Relative Square Error. The formulae of the metrics are given below.

1. Mean Absolute Error (MAE)

$$MAE = \frac{\sum_{i=1}^n |\hat{y}_i - y_i|}{n} \quad (13)$$

2. Root Mean Square Error (RMSE)

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (14)$$

3. Correlation Coefficient (Corr.)

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (15)$$

4. Relative Absolute Error (RAE)

$$\delta = \left| \frac{y_i - \hat{y}_i}{\hat{y}_i} \right| \times 100\% \quad (16)$$

5. Root Relative Squared Error (RRSE)

$$E = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}} \quad (17)$$

6. R-Squared

$$R^2 = 1 - \frac{RSS}{TSS} \quad (18)$$

Here, R = coefficient of determination,
RSS = Sum of squares of residuals,
TSS = Total Sum of squares.
where, y_i = True value of the variable
 $\hat{y}_i$ = Expected value of the variable
 $\bar{y}$ = Mean value of the variable

2.3. Training, validation and test data

We divide the data into three parts viz. Training Data, Validation Data, and Test Data. They are split in a standard way.

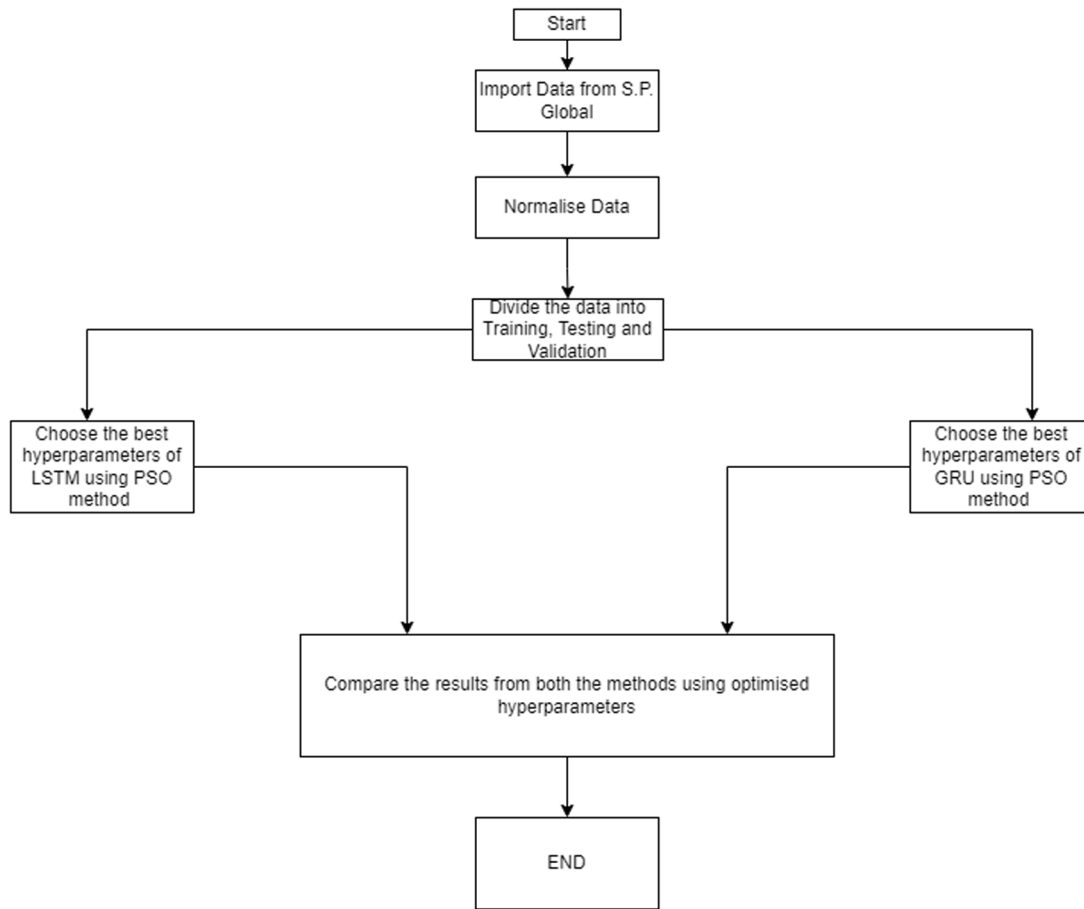


Fig. 5. Flowchart for the entire process.

1. Training Dataset (72%). It is used to build the model.
2. Validation Dataset (8%). It is used to finetune the model using loss, accuracy, etc.
3. Testing Dataset (20%). It is used to test the model. This dataset is kept intact while building the model.

2.4. Normalization of data

In order to build the model, we normalize the data in a value between 0 and 1. The equation for normalization is given below.

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (19)$$

The flowchart for the entire process is given in Fig. 5.

3. Results and discussions

The study aimed to evaluate the effectiveness of LSTM and GRU deep learning models in predicting crude oil prices, which is crucial for decision-making in the energy industry. The models were trained and tested using historical crude oil price data, and Particle Swarm Optimization was used to optimize the models' hyperparameters for improved accuracy and efficiency. The study aimed to compare the performance of LSTM and GRU models and highlight the advantages of the GRU model's mechanism in capturing long-term dependencies. The research also demonstrated how the hybridization of PSO with LSTM and GRU can improve optimization efficiency. The study's findings can be valuable for decision-making in industries such as energy, finance, and economics, demonstrating the effectiveness of deep learning models and optimization techniques in forecasting time-series data. The study focused on capturing profitable trades for short time periods,

typically in a day or two. To achieve this goal, the analysis was limited to five lagged days to determine the influence of price movements on the next day's trade. This approach was specifically designed to cater to option traders who may be interested in trading weekly Crude Oil option expiries. By analyzing the previous five days' price movements, traders can make informed decisions about the direction of future price movements and take advantage of profitable trading opportunities. These findings can be valuable for option traders looking to capitalize on short-term price movements in the Crude Oil market. Therefore, the study spent a lot of experiments to explore the effect of lagged days settings on the accuracy of crude oil forecasting.

In this section, we discuss the results obtained from the algorithms.

In the beginning the presence of any trends in the data is checked. The additive trends and the multiplicative trends for the oil data are checked and are given in the Figs. 6 and 7. It is evident from the figure that there are no additive or multiplicative trends. We compute the results for the unoptimized case. Then, we compute the results for the optimized values of the different hyperparameters.

We will repeat this for lagged days = 3,4,5. For each cases, we will show the "training vs. validation" loss graph, "Forecasted Value vs. Original Value", and "Magnitude of Error" graph. The "Magnitude of Error" is found out by subtracting original value from the forecasted value for the test data. The results are given in Tables 2, 3, 4, 5, 6, 7; and the figures are given from Tables 8 to 13.

Five important hyperparameters are to be optimized. The optimized hyperparameters are given in Tables 3, 5 and 7 for both LSTM and GRU methods. We have chosen six performance metrics. The results of these performance metrics are given in Tables 2, 4 and 6. The results of the unoptimized LSTM and GRU models are given. The results so obtained are for a default set of values for hyperparameters. The default values of the hyperparameters are given below.



Fig. 6. Additive Trends.



Fig. 7. Multiplicative Trends.

Table 2

Different measure of errors of the two optimized methods for crude oil when the number of lagged days taken into account is 3 days.

Algorithms	MAE	RMSE	Corr.	RAE	RRSE	R-squared
LSTM	0.918	1.26	0.990	0.07	0.07	0.9855
GRU	1.000	1.34	0.997	0.08	0.08	0.9936
PSO-LSTM	0.913	1.25	0.996	0.07	0.07	0.9937
PSO-GRU	0.925	1.27	0.997	0.08	0.08	0.9936

Table 3

Optimized values for different hyper parameters when the number of lagged days taken into account is 3 days.

Algorithms	Gradient descent algorithm	Activation	Loss	Batch size	Epochs
PSO-LSTM	nadam	sigmoid	mae	18	85
PSO-GRU	nadam	tanh	mse	10	16

Table 4

Different measure of errors of the two optimized methods for crude oil when the number of lagged days taken into account is 4 days.

Algorithms	MAE	RMSE	Corr.	RAE	RRSE	R-squared
LSTM	1.00	1.354	0.9966	0.085	0.085	0.9894
GRU	0.92	1.24	0.9965	0.083	0.079	0.9937
PSO-LSTM	0.93	1.278	0.9969	0.080	0.080	0.9935
PSO-GRU	0.91	1.23	0.9970	0.078	0.078	0.9939

Table 5

Optimized values for different hyper parameters when the number of lagged days taken into account is 4 days.

Algorithms	Gradient descent algorithm	Activation	Loss	Batch size	Epochs
PSO-LSTM	adam	tanh	mse	32	51
PSO-GRU	nadam	tanh	mse	32	90

Table 6

Different measure of errors of the two optimized methods for crude oil when the number of lagged days taken into account is 5 days.

Algorithms	MAE	RMSE	Corr.	RAE	RRSE	R-squared
LSTM	1.187	1.572	0.993	0.100	0.100	0.9860
GRU	0.955	1.283	0.991	0.082	0.082	0.9931
PSO-LSTM	1.171	1.571	0.996	0.099	0.099	0.9900
PSO-GRU	0.939	1.282	0.996	0.081	0.081	0.9934

Table 7

Optimized values for different hyper parameters when the number of lagged days taken into account is 5 days.

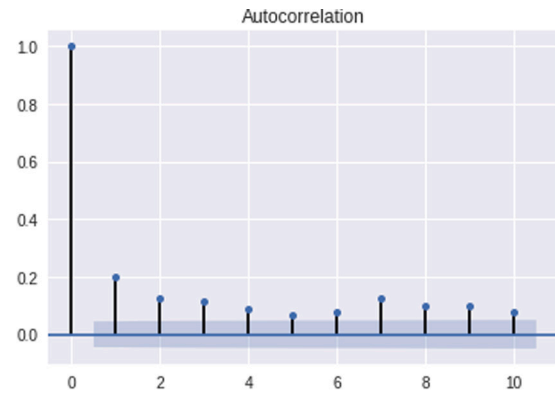
Algorithms	Gradient descent algorithm	Activation	Loss	Batch size	Epochs
PSO-LSTM	nadam	tanh	mse	40	29
PSO-GRU	nadam	tanh	mae	28	71

- Gradient Descent Algorithm = tanh
- optimizer = adam
- loss = maximum absolute error
- epochs = 50
- batch size = 28

We could see that the application of Particle Swarm Optimization on the hyperparameters of LSTM and GRU models improved the results of the score metrics. It is worth mentioning here that we did not use any denoising method, so as to keep the noise terms. That is because the extreme values, noise terms, etc. can be potential opportunities to make successful trades.

We could see that the best algorithm is the PSO-GRU algorithm, taking 4 lagged days, with hyperparameters given below.

- Gradient Descent Algorithm = nadam

**Fig. 8.** Autocorrelation of error terms.

- Activation = tanh
- Loss = mse
- Batch Size = 32
- Epochs = 90

After performing the chi-square test on the test dataset and the expected values of the test dataset, the chi-square value of 67.37 is obtained. For 1000 degrees of freedom, the critical chi-square value is 1074.679 for a 5% level of significance which means that the probability of getting a result at least as extreme as 1074.679 is 5%. In our case, the degree of freedom is 1944 and the chi-square value is $67.37 < 1074.679$. So we accept our null hypothesis that the expected value is correct and it is not some whimsical prediction.

The Autocorrelation test for error terms is then calculated, and the diagram for the first 10 lag points is given in Fig. 8. We then performed the Durbin Watson test and found the value to be 1.6.

Table 14 compares the RMSE values of four different models with our present model. The models used by other authors are Support Vector Machine (SVM), Artificial Neural Network (ANN), Ensemble Empirical Mode Decomposition (EEMD), Particle Swarm Optimization (PSO), and Generalized Autoregressive Conditional Heteroskedasticity (GARCH). The results analysis of Table 14 showed that our PSO-GRU model outperformed all other models in terms of RMSE. The RMSE value for PSO-GRU was 1.23, while the other models had RMSE values ranging from 1.821 to 2.04. This suggests that PSO-GRU is more accurate at forecasting crude oil prices than these other models. The dataset used for the deep learning models in this study consisted of crude oil prices, which remained consistent across all models. The dataset used in this present study was the largest of all, covering the last four decades for analysis. In contrast, other datasets analyzed in the study were based on smaller time periods. The larger dataset used in this present study allowed for a more comprehensive analysis of crude oil prices over a longer time frame, providing a more accurate picture of price trends and patterns. Therefore, the difference in testing periods highlighted the importance of using a comprehensive dataset for accurate forecasting and decision-making in the energy industry.

4. Conclusions

Deep Learning is one of the latest techniques to predict time-series data. In deep learning techniques, LSTM and GRU are one of the best techniques to forecast data. In this research, hybrid PSO-LSTM and PSO-GRU methods are used to examine the utility of LSTM and GRU methods in predicting crude oil prices. The important points obtained from this research are mentioned below.

Table 8
3 lagged days experimental graphs of a sample and a hyperparameter optimized using LSTM.

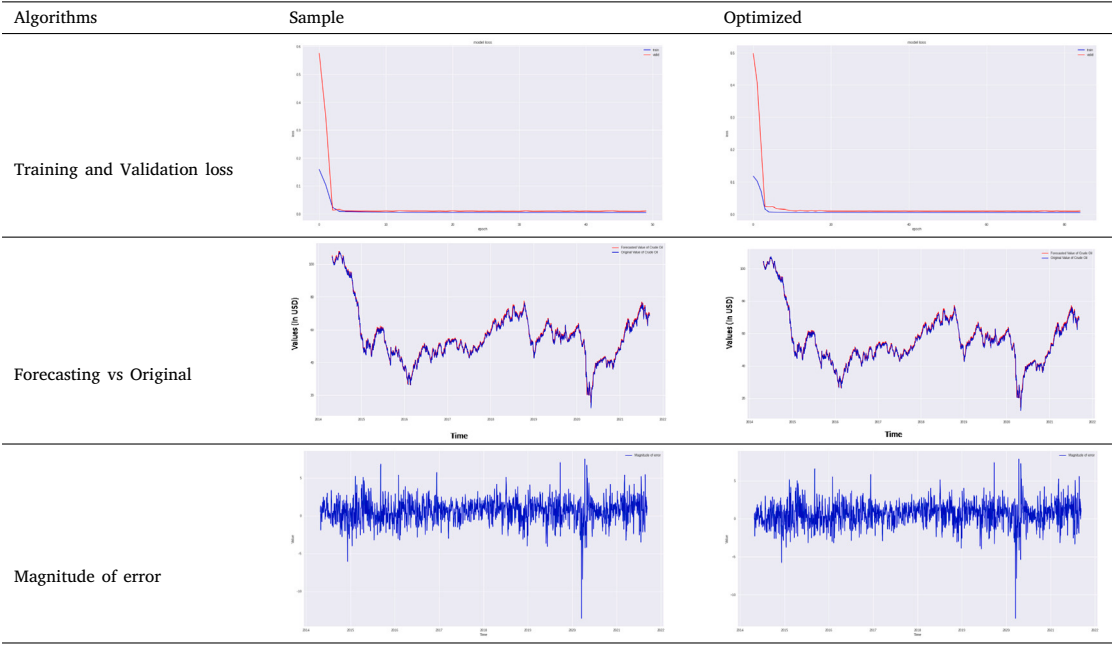
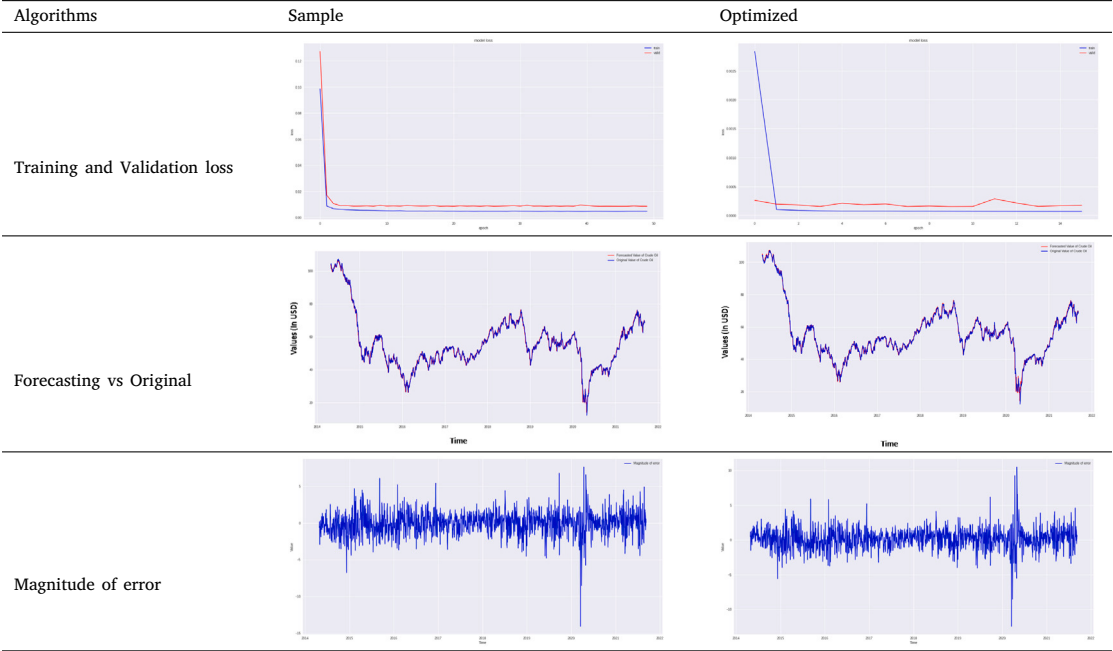


Table 9
3 lagged days experimental graphs of a sample and an hyperparameter optimized using GRU.



- The prices of crude oil can indeed be forecasted with a RMSE of 1.23 and an R-squared value of 0.9939.
- LSTM and GRU methods help forecast crude oil prices. Compared to LSTM, the GRU is more beneficial to forecast crude oil prices.
- When PSO is hybridized with LSTM and GRU, the efficiency of the model increases. Thus PSO is an efficient method to increase the efficiency of the hyperparameters of LSTM and GRU methods.

Table 10
4 lagged days experimental graphs of a sample and a hyperparameter optimized using LSTM.

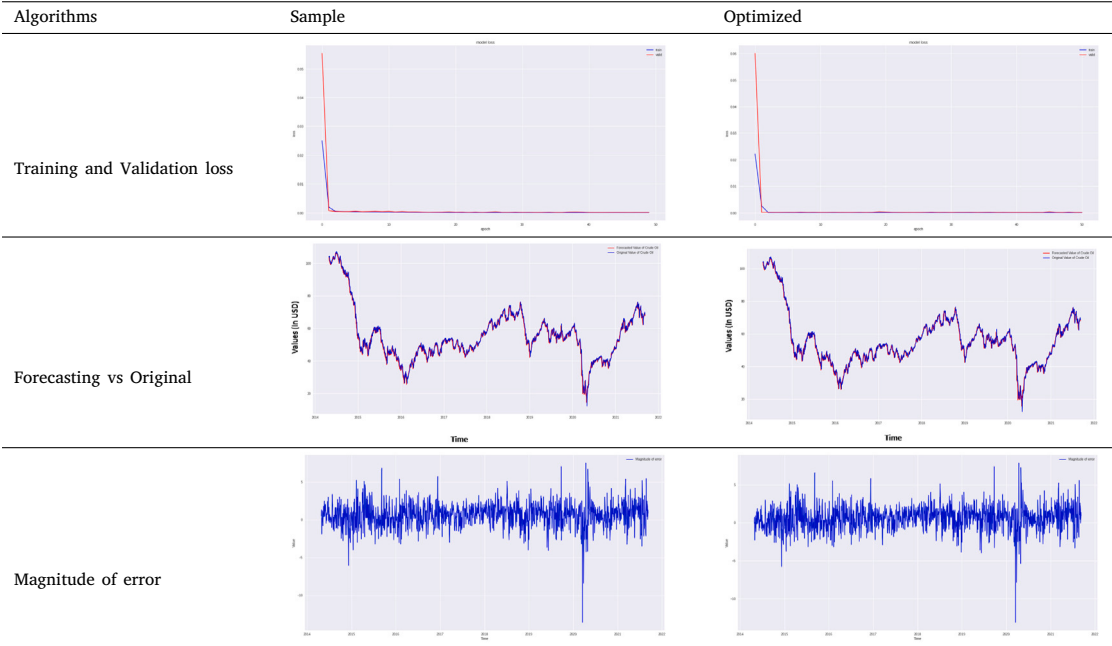
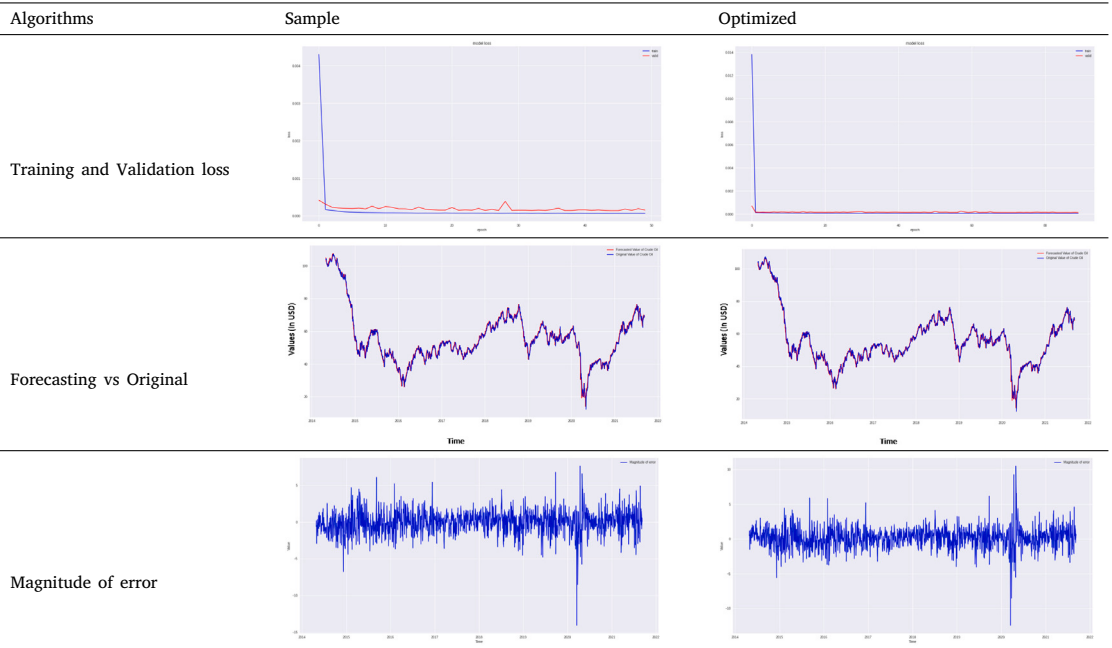


Table 11
4 lagged days experimental graphs of a sample and a hyperparameter optimized using GRU.



Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

CRedit authorship contribution statement

Abhibasu Sen: Conception and design of study, Acquisition of data, Analysis and/or interpretation of data, Writing – original draft, Writing – review & editing. **Karabi Dutta Choudhury:** Conception and design of study, Analysis and/or interpretation of data, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

All authors approved the version of the manuscript to be published.

Table 12
5 lagged days experimental graphs of a sample and a hyperparameter optimized using LSTM.

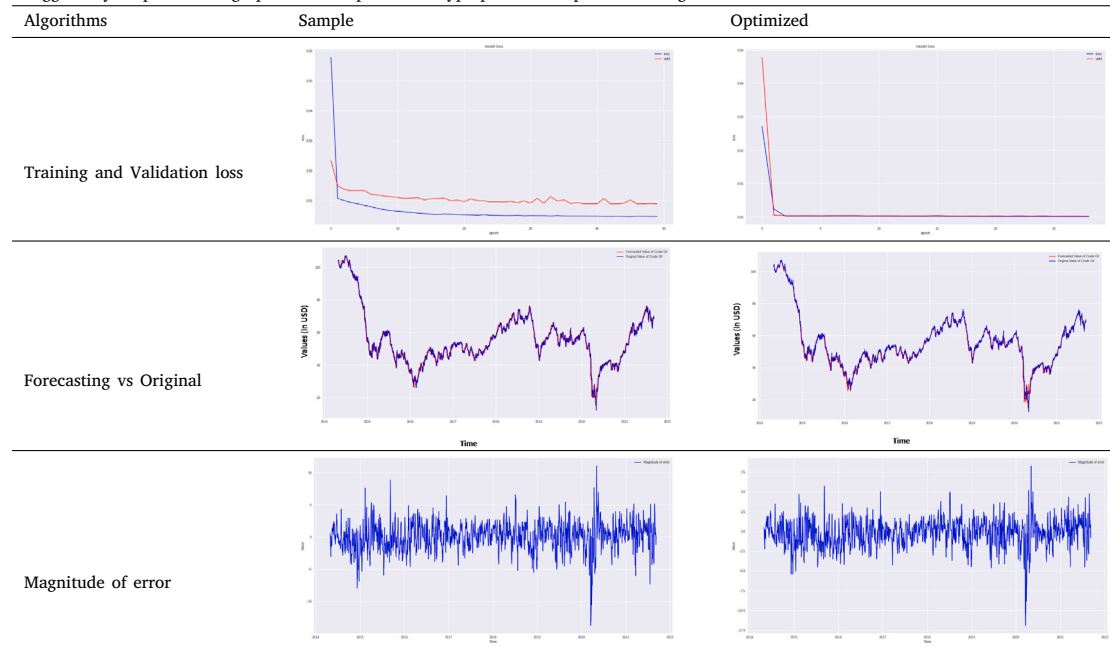


Table 13
5 lagged days experimental graphs of a sample and a hyperparameter optimized using GRU.

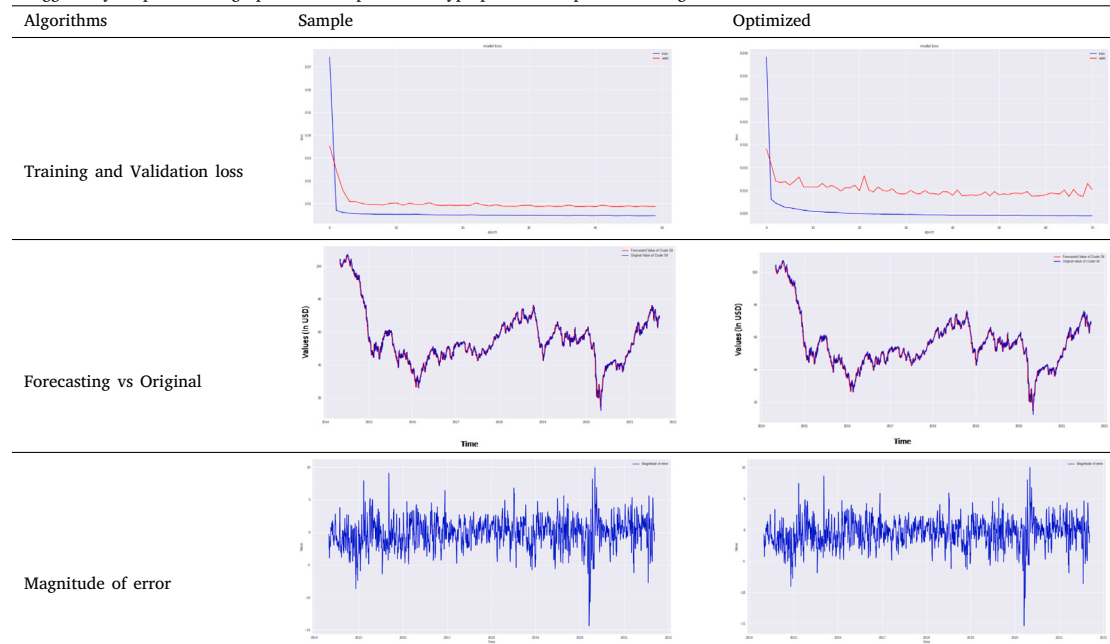


Table 14
A comparison wrt. RMSE of four different models with our model.

Author	RMSE	Models used
Xie et al. (2006b)	1.8210	Support Vector Machine
Moshiri and Foroutan (2006)	2.04	Artificial Neural Network
Zhang et al. (2015)	1.97	Ensemble Empirical Mode Decomposition, ensemble empirical mode decomposition, particle swarm optimization, and generalized autoregressive conditional heteroskedasticity
Jammazi and Aloui (2012b)	1.894	Harr A trous wavelet decomposition multilayer back propagation neural network
Our work	1.23	PSO-GRU

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